

$$\begin{cases} \mathrm{d}X_t^i &= \mathrm{Pev}(\theta_t^i)\mathrm{d}t + \sqrt{2\sigma_x}\mathrm{d}W_t^i \\ \mathrm{d}\theta_t^i &= \chi n(\theta_t^i) \cdot \frac{1}{N} \sum_{j \neq i} \nabla K(X_t^i + \tau v(\theta_t^i) - X_t^j) + \sqrt{2}\mathrm{d}B_t^i \end{cases}$$

Well-posedness

° Singular (Newtonian) $\nabla K: |\nabla K(x)| \sim |x|^{-1}$ (dW, thesis, 2025)

$$\circ \quad \mathrm{d}(X_t^i + \tau v(\theta_t^i) - X_t^i) = \mathrm{Pe}[v(\theta_t^i) - v(\theta_t^j)]\mathrm{d}t + \tau F\left(\theta_t^i, |X_t^i + \tau v(\theta_t^i) - X_t^j|_{j \neq i}^{-1}\right)\mathrm{d}t + \tilde{N}(\theta_t^i)\mathrm{d}\tilde{W}_t$$

$$^{\circ} \text{ For } \sigma_x > 0, \text{ if } \tau \text{ small enough, } \mathbb{E} \left[\int_0^T \left| X_t^i + \tau v(\theta_t^i) - X_t^i \right|^{-2+\epsilon} dt \right] \leq C(T)$$

Analysis

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$$\begin{cases} dX_t^i &= \text{Pe} v(\theta_t^i) dt + \sqrt{2\sigma_x} dW_t^i \\ d\theta_t^i &= \chi n(\theta_t^i) \cdot \frac{1}{N} \sum_{j \neq i} \nabla K(X_t^i + \tau v(\theta_t^i) - X_t^j) + \sqrt{2} dB_t^i \end{cases}$$

- Well-posedness

- Singular (Newtonian) ∇K : $|\nabla K(x)| \sim |x|^{-1}$ (dW, thesis, 2025)

- $d(X_t^i + \tau v(\theta_t^i) - X_t^j) = \text{Pe}[v(\theta_t^i) - v(\theta_t^j)]dt + \tau F\left(\theta_t^i, |X_t^i + \tau v(\theta_t^i) - X_t^j|_{j \neq i}^{-1}\right) dt + \tilde{N}(\theta_t^i) d\tilde{W}_t$

- For $\sigma_x > 0$, if τ **small enough**, $\mathbb{E} \left[\int_0^T \left| X_t^i + \tau v(\theta_t^i) - X_t^i \right|^{-2+\epsilon} dt \right] \leq C(T)$

Analysis

$$\begin{cases} dX_t^i &= \text{Pe} v(\theta_t^i) dt + \sqrt{2\sigma_x} dW_t^i \\ d\theta_t^i &= \chi n(\theta_t^i) \cdot \frac{1}{N} \sum_{j \neq i} \nabla K(X_t^i + \tau v(\theta_t^i) - X_t^j) + \sqrt{2} dB_t^i \end{cases}$$

- Mean-field limit PDE

$$\begin{cases} \partial_t f &= \sigma_x \Delta_x f - \text{Pe} \nabla_x \cdot (v(\theta) f) + \partial_\theta^2 f - \chi \partial_\theta (n(\theta) \cdot (\nabla K * \rho)_\tau f) \\ f(t=0) &= f_0 \end{cases}$$

- Notation:

- $g_\tau(x, \theta) = g(x + \tau v(\theta))$
- $f * g(x) = \int_{\mathbb{T}^2} f(x - y) g(y) dy$
- $\rho(x) = \int_0^{2\pi} f(x, \theta) d\theta$